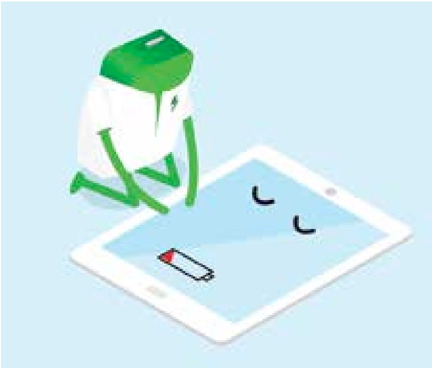
3 floors a day
A chinese businessman built a 57 storey building in 19 days. And that's not all he plans on doing. Read on: **http://dgit.in/57storey**

Who gaurds the guards?
One of the world's leading Anti-Virus company, Kaspersky was the target of a recent sophisticated cyber attack. **http://dgit.in/1MKUUid**



Our devices are more power-hungry

- **Designer Carbon**

In their attempts to improve battery life and performance, scientists have come up with a new pattern of carbon electrodes that is said to boost a battery's performance. The 'designer carbon', that's versatile and controllable, has been developed by scientists at Stanford University. and is said to have exceptional energy-storage capacity, which will take the performance of the lithium-sulphur and other batteries to a whole new level.

Made by heating coconut shells to a very high temperature and then chemically treating the result, this new activated carbon has pores and holes in it leading to greater surface area to store more charges and catalysing more reactions.

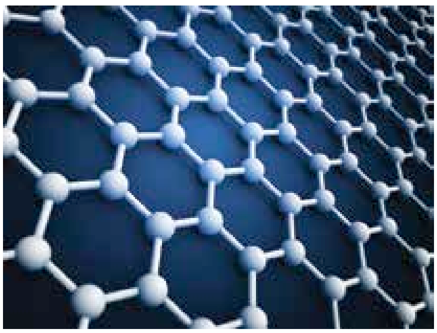
The only drawback of this 'designer carbon' is the limited connectivity between pores, which may limit its ability to conduct electricity. However, it's gaining more popularity with each passing day as people are trying to use them to create new batteries with high power density.

Lithium sulphur batteries are said to benefit the most from the introduction of these electrodes which use chemical reactions between lithium and sulphur to produce power,are lighter than lithium-ion batteries and much more power efficient.

- **Metal-Air batteries**

These work due to the reaction between metal and oxygen and use a porous carbon electrode to trap the ions and catalyse the reaction. Their efficiency depends on the metal used and its reaction with oxygen. Some such batteries are:

1. **Lithium oxygen battery:** These batteries are lighter in weight compared to the lithium-ion batteries since the oxygen used is taken from the environment and need not be stored in the battery. These batteries provide greater power density due to the porous carbon cathode.
   The produced lithium oxygen compound deposits on the carbon but doesn't form a lattice with it, making it easier to recharge.
   Its drawbacks are the use of lithium itself. As Li is extremely reactive, it must not react with the electrolyte while in contact with it. Also H_2O and CO_2 enter the battery along with Oxygen and form compounds that prevent regeneration of Li on recharging.
2. **Sodium air battery:** These batteries use sodium metal, and are more efficient than there lithium counterparts, but are less energy dense. They can be recharged to 93% efficiency and are cheaper than the lithium air batteries. Lithium, being highly reactive, converts to Li_2O_2 on reaction with oxygen which makes it harder to recharge it. Sodium, on the other hand, forms Na_2O as the major product on a nanoscale, which is easier to decompose, thus making it more efficient
3. **Aluminium oxygen battery:** Some people claim that aluminium air batteries are the ultimate alternative to the lithium-ion battery as aluminium is cheaper, lighter and less reactive. Also, aluminium produces three electrons on ionising as compared to lithium, which produces only one electron, giving it higher charge capacity. However, because of low discharge voltage the battery possesses a shorter lifetime than lithium batteries.



What is the best material to make batteries

- **Graphene wraps**

By wrapping sulphur with a thin flexible sheet of graphene, scientists are trying to the battery of the near future. Graphene, with its high conductivity, increases the speed of electron transfer in the batteries, thus increasing its capacity.

- **Microbatteries**

Microbatteries, as their name suggests, are very tiny batteries and the development of these microbatteries is another growing field.

Scientists at the University of Illinois at Urbana have created a microbattery that could charge 1,000 times faster than a regular battery and, as claimed, can fit in and power a credit card thin device.

- **Semi-liquid battery**

Considered semi-liquid due to the liquid ferrocene electrolyte, liquid cathode and a solid lithium anode, this battery, developed by researchers at the University of Texas at Austin, has working voltage similar to that of a li-ion battery. Its power density is comparable to that of a supercapacitor, has good energy density, high capacity and a capacity retention of 80%.

The reason for its high power density is said to be the high speed of ions in a liquid medium as well as the increased speed of electrons.

Power-efficient devices

Besides trying to improve the batteries, devices that use less power (thus increasing their battery life) are also being researched. Technologies that can make your device last longer can already be found around us.

- **Ineda Systems**

A new chip, created by the startup company Ineda Systems, works alongside the main processor in a device to perform various functions while allowing the main processor to spend much of its time powered down, thus saving energy. This chip has two or three cores, one of the which always operates with low power consumption The other core/s will consume more power, but will switch on only for heavier tasks.